

Deliverable 2: Data Collection Methodology

Name: Tam Wing Ching

Student ID: 21007532

Project Title: Logistics Performance & Customer Satisfaction Analysis

Dataset: Brazilian E-Commerce Public Dataset by Olist

1. Data Collection Methodology

The data collection process follows a hybrid model consisting of officially documented feedback loops and inferred automated transactional logging:

- **Active Feedback Loop (Official Mechanism):** As verified by Olist, a satisfaction survey is systematically triggered via email once an order is flagged as "delivered" in the database. This active collection method links qualitative consumer sentiment (review_score and review_comment) directly to specific order_id records, enabling correlation analysis between logistics speed and customer satisfaction.
- **Automated Event-Driven Logging (Inferred):** Given the dataset's multi-dimensional timestamps, it is inferred that Olist utilizes an event-driven architecture. Each milestone—from order_purchase_timestamp to order_delivered_customer_date—is captured via API integrations between marketplaces, seller management systems, and third-party logistics (3PL) providers (e.g., Brazilian Correios). This ensures a real-time, digital "paper trail" for every order.
- **Data Extraction & Relational Logic:** The dataset was likely extracted from a production Relational Database Management System (RDBMS). The extraction logic identifies data points through primary keys (order_id, customer_id) and foreign keys (product_id, seller_id), allowing the integration of disparate information—such as payment types, product dimensions, and geographic coordinates—into a unified analytical snapshot.
- **Passive Operational Capture:** The dataset serves as a secondary "operational byproduct." Rather than being manually curated for research, the data reflects actual, high-fidelity business workflows. This minimizes human intervention error and ensures the dataset is a representative "snapshot" of a real-world e-commerce ecosystem.

2. Ethical Considerations and Limitations

Ethical Considerations

- **Licensing & Compliance (CC BY-NC-SA 4.0):** This research adheres to the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license, ensuring proper attribution to Olist and strictly non-commercial use of derivative findings.
- **Anonymization & Differential Privacy:** Olist performed rigorous **Data Masking** to protect stakeholder privacy. PII (Personally Identifiable Information) such as names and phone numbers were removed; seller identities were pseudonymized (stores and partners were replaced by House names from "Game of Thrones"), and geographic data is limited to the zip_code_prefix level to prevent individual re-identification.
- **Informed Consent:** Data processing is conducted under the **Informed Consent** framework. By registering an account and accepting the platform's **Terms of Service (ToS)**, users (sellers and customers) explicitly agreed to the collection and analytical use of their transactional data for service optimization and marketplace analytics.

Limitations of the Dataset

- **Self-Selection & Response Bias:** The review_score may suffer from **Polarization Bias**. Customers with extreme experiences (highly positive or highly negative) are more likely to participate, which may under-represent the "silent majority" of neutral/average transactions.
- **Temporal Imbalance:** There is a significant **Data Density Variance**. The window from 2017 to August 2018 is robust, but records from 2016 and late 2018 are extremely sparse. This limits the reliability of long-term time-series forecasting or year-over-year growth analysis.
- **Survival Bias & Data Sparsity:** Missing timestamps for cancelled or "in-transit" orders can lead to **Survival Bias**. If incomplete records are excluded during analysis, logistics performance metrics (like average delivery time) may appear more optimistic than the reality of all order attempts.
- **Potential Publisher Pruning:** As a commercial entity, Olist may have performed **Strategic Pruning**. Extreme outliers or systemic logistics failures might have been filtered prior to the public release to protect brand reputation, potentially skewing the performance baseline.